Deep Learning for Economics and Finance

Random search and Hyperband for DEQN architectures

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Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Roadmap of This Lecture

I. Motivation

- Why architecture matters
- The hyperparameter space
- Limits of manual tuning

II. Search Methods

- Grid vs. random search
- Bayesian optimization
- Hyperband / successive halving
- Method comparison

III. Implementing the Search

- Defining a search space
- Random Search + SHA from scratch
- Tool landscape (footnote)

IV. Example & Wrap-up

- Genz-function approximation
- Best practices
- Connection to DEQNs

Hands-on: `02_NAS_Random_Search_10D.ipynb` (10-D random search)

`03_NAS_RandomSearch_Hyperband.ipynb` (Random Search + SHA)

Learning objectives

By the end of this lecture you should be able to:

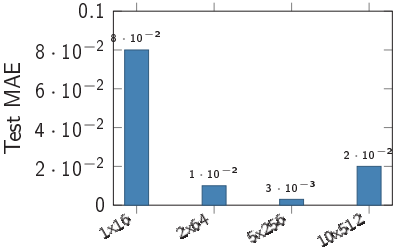
- Diagnose architecture sensitivity and quantify how network depth and width affect approximation error
- Compare grid search, random search, Bayesian optimization, and Hyperband on their computational cost and sample efficiency
- Implement random search and successive halving from scratch in plain Python
- Apply NAS to function approximation problems and choose appropriate search spaces for economic models
- Select production tools (KerasTuner, Optuna, Ray Tune) and apply best practices to scale architecture search efficiently

Why Architecture Matters

Same data, different architectures:

- 1 layer, 16 units $\Rightarrow$ MAE $= 0.08$
- 2 layers, 64 units $\Rightarrow$ MAE $= 0.01$
- 5 layers, 256 units $\Rightarrow$ MAE $= 0.003$
- 10 layers, 512 units $\Rightarrow$ MAE $= 0.02$ (overfit)

Architecture choice can matter more than training duration.



The Hyperparameter Space

Architectural hyperparameters:

- Number of hidden layers $L \in \{1, \ldots, 10\}$
- Units per layer $n_\ell \in \{16, 32, \ldots, 512\}$
- Activation: ReLU, tanh, swish, ...

Training hyperparameters:

- Learning rate $\eta \in [10^{-5}, 10^{-1}]$
- Batch size $B \in \{16, 32, 64, 128\}$
- Epochs, regularization, ...

Combinatorial explosion

Even a modest search space:

$$|\mathcal{H}| = 5 \times 8 \times 3 \times 20 \times 4 = 9{,}600$$

- Each trial: minutes to hours
- Full grid: infeasible

Manual Tuning

The "grad-student descent" approach:

1. Pick an architecture (intuition / paper)
2. Train, evaluate, adjust one thing
3. Repeat until deadline

Problems:

- Doesn't scale beyond a few hyperparameters
- Human bias: we try what we already know
- Not reproducible: results depend on the researcher
- Misses interactions between hyperparameters

Key insight

We need **automated, systematic** methods to search the hyperparameter space efficiently.

Automated Search Strategies

Grid Search

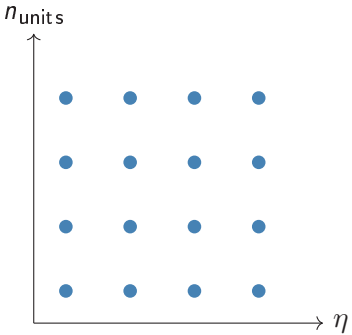
Idea: Enumerate all combinations on a regular grid.

Algorithm

1. Define discrete set for each HP
2. Evaluate *every* combination
3. Return the best

Cost: $\mathcal{O}(k^d)$ — exponential in dimension d.

Pros: Simple, embarrassingly parallel.
Cons: Wastes budget on unimportant dimensions.



16 evaluations

Random Search

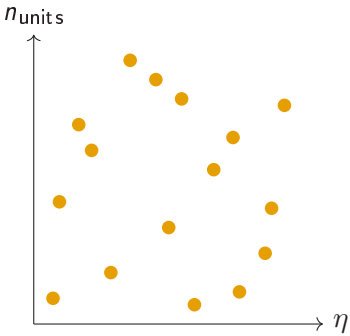
Idea: Sample configurations uniformly at random.

Key result (Bergstra & Bengio, 2012)

When only a *few* dimensions matter, random search finds good configurations with fewer trials than grid search.

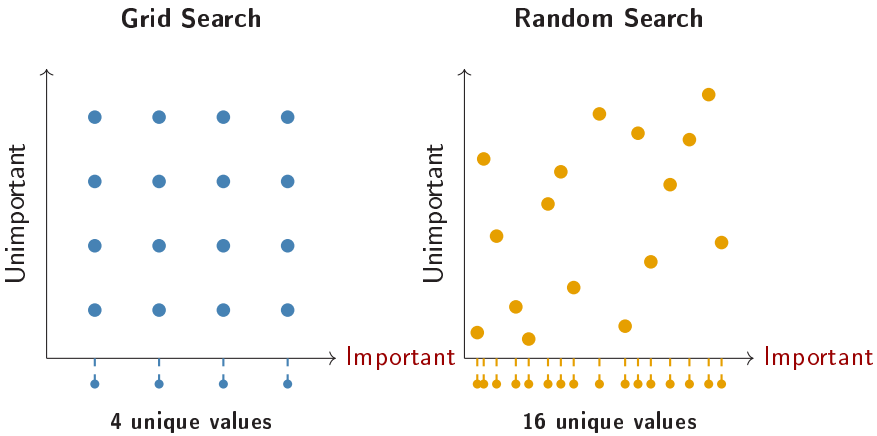
Why? Random points explore each dimension *independently* — more unique values per HP.

Cost: $\mathcal{O}(n)$ — linear in budget n.



16 evaluations

Grid vs. Random — Why Random Wins



When one dimension matters, the grid wastes most evaluations on the unimportant axis; random search samples the important axis more densely.

Bayesian Optimization

Idea: Build a *surrogate model* of $f(\text{HP}) \to \text{loss}$ and use it to choose the next trial *intelligently*.

1. Fit a Gaussian process (GP) to observed trials
2. Compute an acquisition function (e.g., Expected Improvement)
3. Evaluate the HP with highest acquisition value
4. Update the GP and repeat

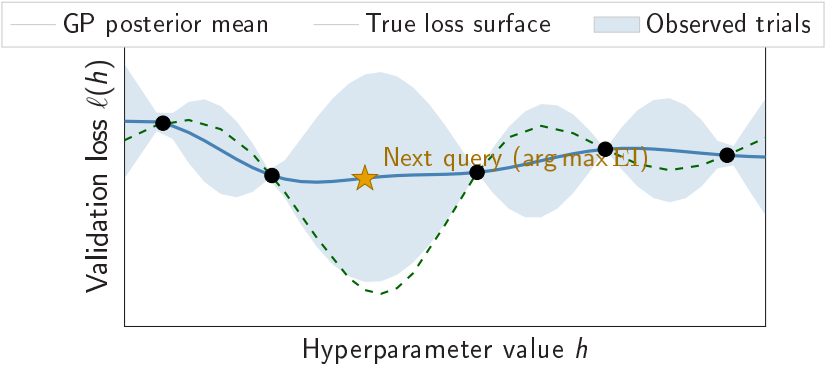
Key advantage: Sample-efficient — exploits structure in the response surface.

Expected Improvement

$$\text{EI}(x) = \mathbb{E}[\max(f^* - f(x), 0)]$$

- f^*: best observed value
- Balances *exploration* (high uncertainty) and *exploitation* (low predicted loss)

Bayesian Optimization — Visual



The GP posterior mean *interpolates* the five observations and is *wide* in unexplored gaps. The Expected-Improvement acquisition function peaks at $h \approx 3.75$, trading off the high posterior variance there against the moderately low predicted mean.

Hyperband

Idea: Start many configurations cheaply; stop the worst early and give more resources to promising ones.

Successive Halving (SHA, $\eta = 3$)

1. Start n random configs with budget b
2. Train each for b epochs, evaluate
3. Keep the top $1/\eta$ of configs (discard the rest)
4. Multiply the budget by η, repeat

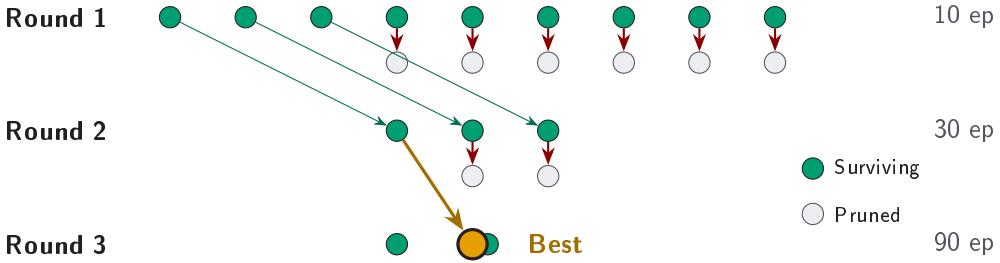
Hyperband = SHA with several brackets (different n/b trade-offs), each bracket itself being a stage cascade like $9 \to 3 \to 1$ above. The usual default is $\eta = 3$.
Resource use: many cheap trials; full budgets only for survivors.

Key insight:

- Bad configs identifiable *early*
- No need for 200 epochs to know $\eta = 10$ diverges
- Exploits correlation between early and late performance

Li et al. (2018): large speedups when early and late performance are correlated.

Hyperband — Successive Halving



$9 \to 3 \to 1$ (factor $\eta = 3$): total cost $= 9 \times 10 + 3 \times 30 + 1 \times 90 = 270$ epoch-equivalents (vs. $9 \times 90 = 810$ for full training).

Bracket vs. schedule. This $9 \to 3 \to 1$ cascade is one SHA bracket. Hyperband itself runs several brackets in parallel, each with a different starting (n, b) trade-off, hedging across the unknown correlation between early- and late-training performance.

Method Comparison

Property	Grid	Random	Bayesian	Hyperband
Computational cost	$\mathcal{O}(k^d)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(n \log n)$
Sample efficiency	Low	Medium	High	High
Parallelisable	Fully	Fully	Limited	Partially
Handles interactions	No	Partially	Yes	No
Early stopping	No	No	No	Yes
Implementation	Trivial	Trivial	Moderate	Moderate

Practical advice:

- **Quick exploration:** Random search (strong baseline)
- **Limited budget:** Bayesian optimization (sample-efficient)
- **Large search space:** Hyperband (aggressive early stopping)

Implementing the Search

Random Search and Successive Halving, from scratch

Defining a Search Space

A plain dict — no library required

```
SPACE = {
    "layers":     [1, 2, 3, 4, 5],
    "units":      [32, 64, 96, 128, 160, 192, 224, 256],
    "activation": ["relu", "tanh", "swish"],
    "lr":         ("log_uniform", 1e-4, 1e-2),
}
def sample_config(rng):
    return {
        "layers":     int(rng.choice(SPACE["layers"])),
        "units":      int(rng.choice(SPACE["units"])),
        "activation": str(rng.choice(SPACE["activation"])),
        "lr":         float(10 ** rng.uniform(-4, -2)),
    }
```

No `hp.Int` / `hp.Float` abstraction — the search space is a dict, the sampler is one function.

Random Search from Scratch

30 trials, single function

```
def random_search(n_trials=30, epochs=80, seed=0):
    rng = np.random.default_rng(seed)
    records = []
    for t in range(n_trials):
        cfg = sample_config(rng)
        model = build_model(cfg)
        h = model.fit(X_tr, y_tr, epochs=epochs,
                      validation_data=(X_val, y_val), verbose=0)
        records.append((cfg, min(h.history["val_loss"])))
    return min(records, key=lambda r: r[1])
```

Independent uniform draws; $\mathcal{O}(N)$ cost, embarrassingly parallel (Bergstra & Bengio, 2012).

Successive Halving from Scratch

$n_0 = 27$, $\eta = 3$, three rounds

```
def successive_halving(n0=27, eta=3, r0=8, seed=0):
    rng = np.random.default_rng(seed)
    survivors = [(sample_config(rng), None) for _ in range(n0)]
    r = r0
    while len(survivors) > 1:
        scored = [(cfg, train_for(cfg, epochs=r)) for cfg, _ in survivors]
        scored.sort(key=lambda s: s[1])           # ascending val loss
        keep   = max(1, len(scored) // eta)
        survivors = scored[:keep]
        r *= eta                                  # 8 -> 24 -> 72 epochs
    return survivors[0]
```

Notebook schedule: 27 candidates @ 8 ep $\rightarrow$ 9 @ 24 ep $\rightarrow$ 3 @ 72 ep $\rightarrow$ 1 winner. This is 648 config-epochs, versus 1500 for 30 random-search trials at 50 epochs each: about 2.3$\times$ less compute while testing a comparable number of architectures.

Production Tooling (footnote)

We just wrote both algorithms in ∼25 lines of Python. Real projects rarely hand-roll the loop; established libraries wrap (and parallelise) the same algorithms behind a uniform API:

Library	Algorithms	Notes
KerasTuner	Random, Bayesian, Hyperband	Tight Keras integration
Optuna	TPE, CMA-ES, Hyperband, NSGA-II	Framework-agnostic; popular default
Ray Tune	All of the above + ASHA, PBT	Distributed by design
Hyperopt	TPE, Random	The original TPE reference
Ax / BoTorch	Bayesian (multi-objective, MOO)	Meta; PyTorch-native GPs
NNI	Random, Bayesian, evolutionary, NAS	Microsoft; full graph-NAS support

Why teach algorithms, not wrappers. APIs change; the search loops (Random, SHA/Hyperband, GP+EI, TPE) persist.

Example: Genz Function

Approximation

Experiment Setup

Target function: Genz Gaussian on $[0,1]^2$

$$f(\mathrm{x}) = \exp\Big(-\sum_{i=1}^{d} c_i^2 (x_i - w_i)^2\Big)$$

Data:

- $d = 2$, uniform random samples
- $n_{\text{train}} = 1{,}000$, $n_{\text{test}} = 2{,}000$
- Loss: MSE, metric: test MAE

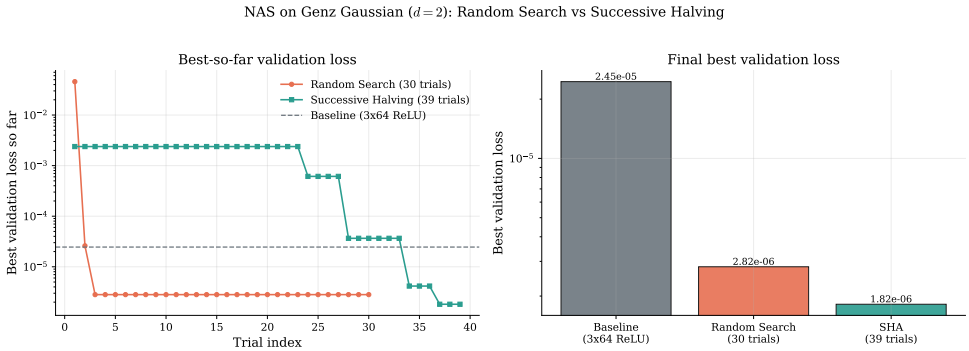
Search space:

- Layers: 1–5
- Units/layer: 32–256 (step 32)
- Activation: ReLU, tanh, swish
- LR: 10^{-4}–10^{-2} (log scale)

Baseline: 3×64 ReLU MLP, $\eta = 10^{-3}$

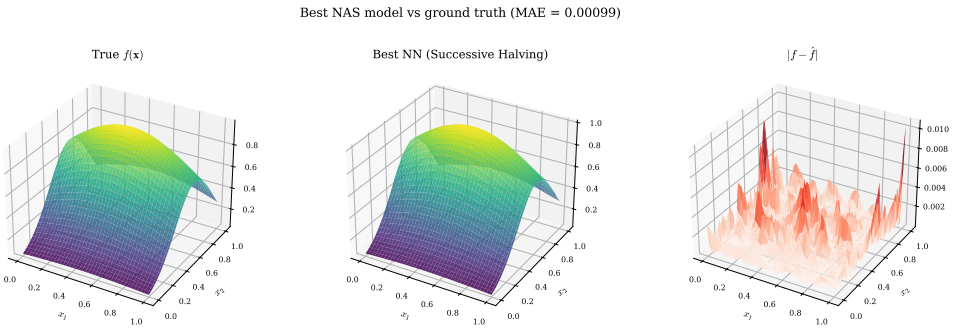
Budget: 30 trials per method

Search Results (Random Search vs. SHA on a 2-D Genz Gaussian)



Generated by `03_NAS_RandomSearch_Hyperband.ipynb`. Random Search uses 30 full trials; SHA uses 27 initial candidates with $\eta = 3$. Both reach MAE of order 10^{-3} from a baseline near $4 \cdot 10^{-3}$; SHA uses about $2.3\times$ less config-epoch budget in this seeded run.

Best Model vs. Ground Truth



Generated by `03_NAS_RandomSearch_Hyperband.ipynb`. Left to right: the true Genz target, the best NAS-found NN approximation, and the absolute error map on $[0, 1]^2$. No placeholder shapes; this is the actual notebook output. The companion `02_NAS_Random_Search_10D.ipynb` offers a 10-D testbed (search loop in plain Python, model in TF/Keras).

Best Practices for NAS

Search space design:

- Start small, expand if needed
- Use log-scale for learning rate
- Constrain to reasonable ranges
- Include the current best as a point

Budget allocation:

- Diminishing returns after $\sim$30 trials
- Hyperband: set `factor=3` (default)
- Use early stopping callbacks

Common pitfalls:

- Search space too large $\Rightarrow$ wasted budget
- Overfitting to validation set $\Rightarrow$ use holdout test
- Ignoring training stability $\Rightarrow$ check learning curves

Rule of thumb

If manual tuning took N trials, automated search with $3N$ trials will likely find something better.

Connection to Economics

Where NAS matters in computational economics:

1. **Surrogate models:** Replace expensive simulations (e.g., climate IAMs) with NNs — architecture determines approximation quality.
2. **DSGE policy functions:** Deep equilibrium nets (Lecture 02) use NNs to approximate policy/value functions. Architecture search over the policy network can improve convergence.
3. **High-dimensional problems:** As d grows (many state variables), the architecture must scale. NAS automates this adaptation.
4. **Reproducibility:** Automated search documents *which* architectures were tried, making results more transparent than "we chose 3 layers with 64 units."

Summary

Key takeaways:

1. Architecture choice strongly affects NN performance
2. Random search is a strong baseline — always try it first
3. Bayesian optimization is most sample-efficient
4. Hyperband / SHA offers the best speed/quality trade-off
5. Production: KerasTuner, Optuna, Ray Tune, Ax all wrap these

References:

- Elsken, Metzen & Hutter (2019) — *NAS: A Survey*, JMLR 20(55); see `readings/links_by_lecture/lecture_05_B04`
- Bergstra & Bengio (2012). *Random search.* JMLR 13.
- Li et al. (2018). *Hyperband.* JMLR 18(185).
- Garnett (2023). *Bayesian optimization.* CUP.
- Snoek et al. (2012). *Practical Bayesian optimization.* NeurIPS.